

Alg. Top: idea attach to top space invariants which can then distinguish spaces

ex: (1) Euler - Poincare' characteristics (i.e. numbers)

(2) Homotopy groups (loop/punctures)

(3) Homology groups (simplicial structure)

$\Rightarrow$ For topological spaces: homeomorphisms $\sim$ equivalence relation on the class of top spaces

$\Rightarrow$ Homotopy: generates equivalence classes that contain continuous maps

let X and Y be spaces

$$\text{Map}(X, Y) = \{ \psi: X \rightarrow Y \mid \psi \in \text{Continuous} \}$$

For $\alpha \in \text{Map}(X, Y)$, $\beta \in \text{Map}(X, Y)$

$\alpha \approx \beta$ (Homotopic)

if $\exists F: [0, 1] \times X \rightarrow Y$ s.t. $\forall x \in X$

i) $F(0, x) = \alpha(x)$

ii) $F(1, x) = \beta(x)$

F : Homotopy from α to β



* Thus: Homotopy divides $\text{Map}(X, Y)$ up into (Homotopic) equivalence classes

$$[X, Y] = \{ [\alpha] \}$$

* If we view $[X, Y]$ as a functor of Y with X fixed then the collection of equivalence classes is a top. invariant of Y

$\Rightarrow$ the use of a homo, h , to change Y for a homeo copy Y' induces a canonical rep of F from α to β

by

$h \circ F$

$$[0,1] \times X \xrightarrow{F} Y \xrightarrow{h} Y'$$

$$\begin{array}{ccc} F(0,x) & X & \xrightarrow{\alpha} Y \xrightarrow{h} Y' \\ F \downarrow & & \\ F(1,x) & X & \xrightarrow{\beta} Y \end{array}$$

$\Rightarrow$ Replaces F from α to β with $F \circ h$ from $h \circ \alpha$ and $h \circ \beta$

$$\begin{array}{ccc} & \beta & \\ F \uparrow \uparrow & & \\ & \alpha & \end{array}$$

$h \circ \beta$
 $h \circ \alpha$

* As a top invariant: $[X, Y] \sim [X, Y']$ for homeomorphic spaces Y and Y'

• Homotopy type: A space X is of the ^{same} homotopy type as Y if $\exists$ a continuous map α

$$\alpha: X \rightarrow Y$$

such that α , though not possessing an inverse, must have a homotopic inverse, say β .

i.e. $\alpha: X \rightarrow Y$

$$\beta: Y \rightarrow X$$

$$\alpha \circ \beta \cong I_Y$$

$$\beta \circ \alpha \cong I_X$$

$$X \simeq Y$$

$\Rightarrow X$ is homotopic equivalent to Y

* Homotopic eq. spaces are those which can be continuously deformed into one another

• Top invariants where one of X and Y are fixed and the other varies are known as invariants of homotopy type

ex: Y and Y' from above are the same homotopy type

ex: $\mathbb{R}^n$ can be contracted to a point, so is of the same

These are homotopy maps i.e. $\exists F_x$

$$F_x: [0,1] \times X \rightarrow X$$

s.t.

$$\begin{cases} F(0,x) = \beta \circ \alpha \\ F(1,x) = I_X \end{cases}$$

$$\begin{cases} F(0,x) = \beta \circ \alpha \\ F(1,x) = I_X \end{cases}$$

same for $\alpha \circ \beta$

homotopy type as a pt.: trivial homotopy/homotopy groups

- But $\mathbb{R}^n$ is not homeomorphic to $\{1\}$

$$\mathbb{R}^n \cong \mathbb{R}^m \text{ iff } n=m$$

So $\dim(X)$ is an example of a top. invariant which is not an invariant of homotopy type.

* For $[X, Y]$, with fixed X an important choice is $X = S^1$

$\leadsto$ For S^1 , we need to define base points *

$\text{Map}_0(X, Y)$: Base pt of X is mapped to that of Y

$w \in \text{Map}_0(X, Y)$ w : based map

$$\hookrightarrow [X, Y]_0$$

* $\rightarrow$ 
 $[X, Y]$: "free" homotopy class

• A pair $\{X, *\}$ of a space and a base pt is called a pointed space

$\leadsto$ Recall For X, Y and Z $\exists$ a natural bijection

$$i: \text{Map}(X, \text{Map}(Y, Z)) \longrightarrow \text{Map}(X \times Y, Z)$$

$$Y^X \equiv \{ \text{set of all maps } \psi: X \rightarrow Y \}$$

then there are bijections

$$Z^{X \times Y} \cong Z^{Y \times X} \cong (Z^X)^Y$$

$$\text{Proof: } Z^{X \times Y} = \{ \psi: X \times Y \rightarrow Z \}$$

$$(i) \quad a: Z^{X \times Y} \rightarrow Z^{Y \times X}$$

$$a(\psi)(y, x) := \psi(x, y)$$

$$\text{while } a^{-1}: Z^{Y \times X} \rightarrow Z^{X \times Y}$$

$$a^{-1}(\phi)(x, y) := \phi(y, x)$$

$$\Rightarrow (a \circ a^{-1})(\phi)(x, y) = a(\phi(y, x))$$

$$= (a \circ \phi)(y, x) = \phi(x, y)$$

$$\Leftarrow (a^{-1} \circ a)(\psi)(y, x) = (a^{-1} \circ \psi)(y, x) = \psi(y, x)$$

$\therefore a$ is a bijection

$$(ii) \quad Z^{X \times Y} \cong (Z^X)^Y$$

$$(Z^X)^Y = \{ \psi: Y \rightarrow Z^X \}$$

$$\hookrightarrow Z^{X \times Y} \cong (Z^X)^Y$$

$$b(4) \in (Z^X)^Y$$

$$b(4) : Y \rightarrow Z^X$$

$$(b \circ 4)(y) : X \rightarrow Z$$

$$(b \circ 4)(y)[x] := \psi(x, y)$$

$$b(4) := \psi(\cdot, \cdot)$$

$$\psi(\cdot, y) \in Z^X$$

$$b^{-1}(2) \in Z^{X \times Y}$$

$$(b^{-1} \circ 2)(x, y) := 2(x, y) = 2(y)(x)$$

$$2 : Y \rightarrow Z^X$$

$$2(y) : X \rightarrow Z$$

$$2(y)(x) \in Z$$

$$2(y)(x) \equiv 2(x, y)$$

$$(b \circ b^{-1}) \circ 2(x, y) = (b \circ 2)(y)[x] = 2(x, y)$$

$$(b^{-1} \circ b \circ 4)(y)[x] = (b^{-1} \circ 4)(x, y) = \psi(x, y)$$

$\therefore$ These bijections come from defining maps

$$\tilde{2}(y)(x) \sim 2(x, y)$$

i.e.

$$Y \xrightarrow{\tilde{2}} Z^X \xrightarrow{\psi} Z$$

$$\gamma(4) \equiv \psi(x)$$

$$(\gamma \circ \tilde{2})(y) = \gamma(\tilde{2}(y)) = [\tilde{2}(y)](x) \leftrightarrow 2(x, y)$$

Q: How do I know $\tilde{2}$ and 2 exist?

* hint to set up a similar Bijection for pointed spaces

$\leadsto$ Need to set up base point of $X \times Y$

Def: The reduced join of X and Y : (Wedge Sum)

$$X \vee Y := X \sqcup_b Y / \sim \text{ (with both base points identified)}$$

The smash product:

$$X \wedge Y = \frac{X \times Y}{X \vee Y}$$

x_0, y_0

$\leadsto X \vee Y$ is smashed to a point: Becomes the base point of

$$X \vee Y = X \sqcup Y / \sim, \quad x_0 \sim y_0$$

$$X \vee Y \sim X \times \{y_0\} \cup \{x_0\} \times Y$$

(i) Have equivalence (x_0, x) are all same
 (ii) " " (x, y_0) " "

$$\leadsto \bar{X} \wedge \bar{Y} = \bar{X} \times \bar{Y} / \sim$$

Ex: $S^2 \wedge S^1 \cong S^3$

* $\exists$ a bijection i_*

$$i_*: \text{Map}_0(\bar{X}, \text{Map}_0(\bar{Y}, \bar{Z})) \rightarrow \text{Map}_0(\bar{X} \wedge \bar{Y}, \bar{Z})$$

as above.

* $S^1 \wedge X$: suspension or reduced suspension of X

$$S^1 \wedge X = SX$$

$$\bullet \text{Map}_0(S^1, X) \cong \Omega X$$

$$S^1 \wedge S^m = S^{m+1} \cong S^{m+1}$$

* Now look at S and Ω as operators which act on X .

S & Ω are functors

$\leadsto$ Will see that S and Ω are adjoints of one another

Canonical Bijection:

$$\text{Map}_0(SX, \bar{Y}) \cong \text{Map}_0(X, \Omega \bar{Y})$$

which follows from i_*^{-1} and

$$i_*^{-1}: \text{Map}_0(\underbrace{X \wedge S^1}_{SX}, \bar{Y}) \rightarrow \text{Map}_0(X, \underbrace{\text{Map}_0(S^1, \bar{Y})}_{\Omega \bar{Y}})$$

which implies the set of homotopy classes:

$$[SX, \bar{Y}]_0 \cong [X, \Omega \bar{Y}]_0$$

which displays the adjoint nature of S & Ω .

* View group structures in algebraic top as group structures which arise from $[X, Y]_0$. This occurs in two ways

(i) Suppose $Y = \{G, *\}$ is a group, then $[X, Y]_0$

Let $\alpha, \beta \in \text{Map}_0(X, G)$, denote the group product as

$$\alpha \bullet \beta: X \rightarrow G$$

$$x \mapsto \alpha(x) * \beta(x)$$

* The product $\alpha \bullet \beta$ then induces a product in $[X, G]_0$

Let $[\alpha], [\gamma] \in [X, G]_0$

then $[\alpha] \star [\gamma] := [\alpha \bullet \gamma]$

and $([\alpha] \star [\gamma])(x) = \alpha(x) \star \gamma(x)$

$\Rightarrow$ need to show this doesn't depend on the chosen rep

Let $\alpha \sim \beta$ and $\gamma \sim \delta$

$$[\beta] \star [\delta](x) = [\beta \bullet \delta](x) = \beta(x) \star \delta(x)$$

(Not too sure)...

- To obtain a group structure on $[X, G]_0$ it is not nec. that $\star$ on G be a group prod as it is sufficient that it descends to a group prod. on the homotopy classes $[X, G]_0$.
- For $\star: Y \times Y \rightarrow Y$ which leaves $[X, Y]_0$ a group we call Y a homotopy associative H-space.

$[X, Y]_0$: group $\Rightarrow X$: any space
 Y : H-space

* Replace $\star$ with the comultiplication map $\star'$ (H-cospaces)
 $\star': X \rightarrow X \vee X$

as above this induces a product law $\bullet'$ on $[X, Y]_0$
 X : co-Hspace

ex: $X = S^1 \mathbb{Z} = S^1 \wedge \mathbb{Z}$, for any space $\mathbb{Z}$
 so here

Homotopy groups:

$$\pi_1(X) := [S^1, X]_0 = \{[\alpha]\} \quad \left(\begin{array}{l} \text{Fundamental Group / First Homo.} \\ \text{group} \end{array} \right)$$

$\Rightarrow$ Homotopy classes of Based loops on X

• Think of this as embeddings



• Do not distinguish between homotopic loops

Group structure on $\pi_1(X)$

Let $[\alpha], [\beta] \in \pi_1(X)$ i.e. $\alpha, \beta \in \text{Map}(S^1, X) = \Omega X$

$$\alpha * \beta(t) := \begin{cases} \alpha(2t) & , 0 \leq t \leq \frac{1}{2} \\ \beta(2t-1) & , \frac{1}{2} \leq t \leq 1 \end{cases}$$

t : coordinate on unit S^1

* This is simply a concatenation and to ensure this is continuous then one must have $\alpha(1) = \beta(0)$

* Some notes on transitivity & representative independence.

Let $w \in \Omega X$, then the inverse path is given by $w^{-1}(s) = w(1-s)$, $s \in S^1$ $(w^{-1}) : S^1 \rightarrow X$

Transitivity: $\alpha \sim \beta$ and $\beta \sim \gamma$, then $\alpha \sim \gamma$:

$$\begin{aligned} \alpha \sim \beta &\rightarrow F(t, x) \\ \beta \sim \gamma &\rightarrow G(t, x) \\ \alpha \sim \gamma &\text{ with } H(t, x) := \begin{cases} F(2t, x), & 0 \leq t \leq \frac{1}{2} \\ G(2t-1, x), & \frac{1}{2} \leq t \leq 1 \end{cases} \end{aligned}$$

$\uparrow t$

γ

G	β
F	

α

Lemma: $\alpha \sim \alpha' \wedge \beta \sim \beta' \Rightarrow \alpha * \beta \sim \alpha' * \beta'$

$$H(s, x) = \begin{cases} F(2s, x) & , 0 \leq s \leq \frac{1}{2} \\ G(2s-1, x) & , \frac{1}{2} \leq s \leq 1 \end{cases}$$

$$\begin{aligned} \leadsto [\alpha][\beta] &= [\alpha * \beta] \\ \alpha * \beta(t) &= \begin{cases} \alpha(2t) & \dots \\ \beta(2t-1) & \dots \end{cases} \\ [\alpha'][\beta'] &= [\alpha' * \beta'] \\ (\alpha' * \beta')(t) &= \begin{cases} \alpha'(2t) & \dots \\ \beta'(2t-1) & \dots \end{cases} \end{aligned}$$

$$H(0, x) = F(x, 0) = \alpha(0) \quad \checkmark$$

$$H(\frac{1}{2}, x) = F(0, 1) = \alpha(1) =$$

* Higher Homotopy groups: $\pi_n(X) = [S^n, X]_0$

Recall: $\Omega X \equiv \text{Map}_0(S^1, X)$

$$S^1 X \equiv S^1 \wedge X$$

$$\text{And } [S^1 X, Y]_0 \cong [X, \Omega Y]_0$$

which follows from $\text{Map}_0(S^1 X, Y) \cong \text{Map}_0(X, \Omega Y)$

$$S^n = S^1 \wedge S^{n-1} = S S^{n-1} \quad | \quad \Omega X: \text{H-space}$$

$$[SS^{n-1}, X]_0 = [S^{n-1}, \Omega X]_0$$

and so $[S^{n-1}, \Omega X]_0$ is a group and

so $\pi_n(X) = [S^n, X]_0 = [S^{n-1}, \Omega X]_0$ is also a group

• If X is Riemann surface of $g \geq 2$: $\pi_1(X)$ is non-commun.

• $\pi_n(X)$ are always commutative: Since $n \geq 2$

$$\pi_n(X) = [S^n, X]_0 = [S^{n-1}, \Omega X]_0 = [SS^{n-2}, \Omega X]_0$$

which is of the form $[X, Y]_0$, $Y = \Omega X$, $X = SS^{n-1}$

$\xleftarrow{\text{H-space}} \text{co H-space}$
implies comm.

Aside on multiplication:

2 particle state $|A\rangle \otimes |B\rangle$, $|J\rangle = |A\rangle + |B\rangle$

$$\hat{J} \equiv \hat{J}_A \otimes \hat{1}_B + \hat{1}_A \otimes \hat{J}_B \quad (\text{spm rep of } \mathcal{O}(n))$$

$$*': J \rightarrow J \otimes J$$

$$\hat{J} \mapsto \hat{J}_A \otimes \hat{1}_B + \hat{1}_A \otimes \hat{J}_B$$

• For our abstract case $\leftarrow$ Base point.

$$*': X \mapsto (X, x_0)$$

* Co-H space is a pair (X, x_0) with a map $*': X \rightarrow X \vee X$
which makes the diagram commute up to homotopy

$$\begin{array}{ccc} & *' \rightarrow X \vee X & \\ & \downarrow \Delta & \\ X & \xrightarrow{\Delta} X \times X & \end{array} \quad \Delta: x \mapsto (x, x) \quad (\text{diagonal})$$

• Let $X = G$ be a topological group. and let $S^0: x^2 = 1$
 $S^0 \cong \mathbb{Z}_2$ where one of the two are the base pt.

Claim: $\pi_1(G)$ is always commutative

$$\text{Proof: } \pi_1(G) = [S^1, G]_0 = [SS^0, G]_0$$

SS^0 is a co-H-space and G a group $\square$

$$* [X, Y]_0 / \pi_1(Y) \cong [X, Y]$$

$$\pi_m(S^2) = \begin{cases} \mathbb{Z} & m=2 \\ 0 & m>2 \end{cases}$$

For $n \geq 1$,

$$\pi_m(S^n) = \begin{cases} 0 & m < n \\ \mathbb{Z} & m = n \end{cases}$$

$\pi_m(S^n) = \begin{cases} \text{a finite group} & m > n \text{ and } n \text{ odd} \\ \text{"} & \text{"} \end{cases}$

$$\mathbb{Z} \times \mathbb{F}^n, \quad m > n \quad n \text{ even and } m = 2n-1$$

F : finite group

* For $n \geq m+2$ and π_{n+m} :

$$\pi_{m+n}(S^n) = \pi_{m+n+1}(S^{n+1}) = \dots = G_m, \quad \text{say}, \quad n \geq m+2$$

$$G_m = \pi_{2m+2}(S^{m+2}) \quad \text{for } n = m+2$$

Cohomology and homology groups:

Eilenberg-MacLane spaces: spaces w/ only one non-vanishing homotopy group

$K(G, n)$

$$\pi_m(K(G, n)) = \begin{cases} G, & \text{if } m=n \\ 0, & \text{otherwise} \end{cases}$$

ex: $S^1 = K(\mathbb{Z}, 1)$

$\pi_1(S^1) = \mathbb{Z}$

* Given any G and n there always exists some $K(G, n)$

Use these to define cohomology groups

$$H^n(X; G) := [X, K(G, n)]_0 \quad \left(\begin{array}{l} n^{\text{th}} \text{ cohomology group of } X \text{ with} \\ \text{coeff in the group } G \end{array} \right)$$

* $\{[\alpha] \mid \alpha: X \rightarrow K(G, n)\}$

$[\cdot]$: homotopic equivalence

Thus:

$$\begin{cases} \pi_m(X) = [S^m, X]_0 \\ \pi_n(K(G, n)) = [S^n, K(G, n)]_0 = G \\ H^n(S^n; G) = \pi_n(K(G, n)) = G \\ \pi_m(\Omega K(G, n+1)) = \begin{cases} G, & m=n \\ 0, & \text{otherwise} \end{cases} \end{cases}$$

i.e. $\Omega K(G, n+1)$ is also a $K(G, n)$

proof:

$$\begin{aligned} \pi_m(\Omega K(G, n+1)) &= [S^m, \Omega K(G, n+1)] = [SS^m, K(G, n+1)] \\ &= [S^{m+1}, K(G, n+1)] = \pi_{m+1}(K(G, n+1)) = G \end{aligned} \quad \left. \begin{array}{l} m+1=m \\ x \\ m=n \end{array} \right\}$$

$\Omega K(G, n+1) \sim K(G, n)$ and so

$\Omega^m K(G, n+m)$, $m=1, 2, \dots$ are all $K(G, n)$'s

* Since we can express $K \sim \Omega Y$ it is evident

K is an H -space
 H -space: $\sim \Omega Y$
 coH-space: $\sim S\mathbb{X}$

and so $[X, K(G, n)]_0 = H^n(X, G)$ is automatically endowed w/ a group structure.

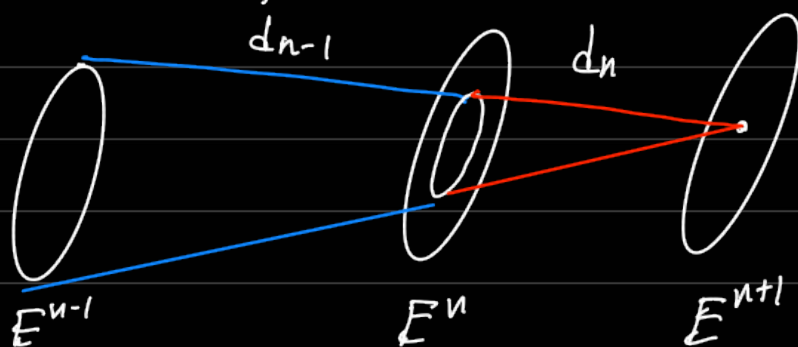
* Complex definition of cohomology groups:

Take a collection of spaces $E^n(X)$ together with maps d_n

$$\dots \xrightarrow{d_{n-2}} E^{n-1}(X) \xrightarrow{d_{n-1}} E^n(X) \xrightarrow{d_n} E^{n+1}(X) \xrightarrow{d_{n+1}} \dots$$

$\{E^n(X), d_n\}$ must have $d_n \circ d_{n-1} = 0 \quad \forall n$

$$\Rightarrow \text{Im}(d_{n-1}) \subseteq \text{Ker}(d_n)$$



If $\text{Im}(d_{n-1}) = \text{Ker}(d_n) \quad \forall n$ then the above is exact

$$H^n_E(X) = \text{Ker}(d_n) / \text{Im}(d_{n-1})$$

H^n are trivial for exact sequences

(u-forms)

ex: (i) Let X be a diff. mfd. d : ext. derivative $E = \Omega^n(X)$

(de Rham)

$H^n(X, \mathbb{R})$:

$$H^n_{\mathbb{R}}(X) = \frac{\text{Ker}(d_n)}{\text{Im}(d_{n-1})} = H^n_{\text{de Rham}}(X) \cong H^n(X; \mathbb{R})$$

$$[X, K(\mathbb{R}, n)]_0$$

(ii) Homology groups $\{E_n, \partial_n\}$: complex dual of $\{E^n, d\}$

$$\dots \xleftarrow{\partial_{n-2}} E_{n-1}(X) \xleftarrow{\partial_{n-1}} E_n(X) \xleftarrow{\partial_n} E_{n+1}(X) \xleftarrow{\partial_{n+1}} \dots$$

$$\partial_{n-1} \circ \partial_n = 0 \quad H_n^E(X) = \text{Ker}(\partial_{n-1}) / \text{Im}(\partial_n)$$

"Boundary of a Boundary is zero"

∂_n : Boundary map

$\{E_n, \partial_n\}$ singular chain complex for X : E_n singular

$$H_n^E(X) \cong H_n(X, \mathbb{Z})$$

n -chains

• Non trivial for $0 \leq n \leq \dim(X)$

cohomology:

$$H^0_{\text{cohom}} = \bigoplus_{i=0}^{\dim X} H^i(X)$$

$H^0(X) = n=0$ $\mathbb{R}$, forms a ring with a cup product $\cup$

• for de Rham $\cup = \wedge$

$$H_{\text{de Rham}}^0(X) \cong H^0(X, \mathbb{R}) = \bigoplus_{n=0} H^n(X, \mathbb{R})$$

$$\cup: H^m(X, \mathbb{R}) \times H^n(X, \mathbb{R}) \rightarrow H^{m+n}(X, \mathbb{R}) \quad | \quad \omega, \nu: \text{closed}$$

$$([\omega], [\nu]) \mapsto [\omega] \cup [\nu] = [\omega \wedge \nu]$$

* $f: N \rightarrow M$, then $\omega \in H^n(M; G)$
 $f^*\omega \in H^n(N; G)$

$$\begin{array}{ccc} H_E^*(N) & \xleftarrow{f^*} & H_E^*(M) \\ \downarrow & & \downarrow \\ N & \xrightarrow{f} & M \end{array}$$

$\therefore f^* H^*(M; G)$ is an object

* Pull-back Bundle: $f^*E \xrightarrow{h} E$
 $\begin{array}{ccc} \pi_f \downarrow & & \downarrow \pi \\ N & \xrightarrow{f} & M \end{array}$

$$f^*E \sim N \times E, \quad f^*E = \{ (p, e) \in N \times E \mid f(p) = \pi(e) \}$$

$$\pi_f(p, e) := p$$

$$h(p, e) := e \quad (\text{projection})$$

* Comm. diagram: $\tilde{e} \in f^*E \Rightarrow (\pi \circ h) = f \circ \pi_f$

* For two homotopic maps $f_1 \simeq f_2$, $f_1, f_2: N \rightarrow M$ then

$f_1^*E \simeq f_2^*E$ are isomorphic

$$f^*E = \bigsqcup_{p \in M} E(f(p)), \quad \text{where} \quad E(x) := \pi^{-1}(x)$$

i.e. $E(x) = E_x \sim \{x\} \times F_x$

Def: A Classifying space: BG

